# Leveraging Few-Shot Learning with Text/Image Conditioned Object Detection for E-Waste Sorting

**Abstract**

Object detection is crucial in industrial applications such as automated waste sorting. Traditional object detection methods rely on extensive data collection, annotation, and model training processes that are often costly. Furthermore, addressing inaccuracies during deployment typically requires additional data collection and model retraining, increasing the overhead. This highlights the need for AI models that can be trained or fine-tuned with minimal data annotation. Moreover, learning new classes or correcting deployment mistakes should be possible without full retraining. Recent advances in transformers have introduced foundation vision-language models that enable performance adjustments through simple text or image prompts. This study explores a few-shot object detection approach for e-waste sorting using the Open-World Vision Transformer (OWL-ViT). Since OWL-ViT is limited to processing a single query per target class, we investigate various strategies to merge multiple queries per class. Our results demonstrate that a fine-grained merging achieves accuracy comparable to conventional closed-set object detectors while eliminating the need for extensive pre-deployment or in-deployment model retraining.

*Keywords:* Waste sorting, Few-shot object detection, Real-time adaptation, Human-in-the-loop systems

## 1. Introduction

Artificial Intelligence (AI) has absorbed considerable attention in various industries, boosting performance while lowering the costs and complexities. However, the use of AI in *circular economy* is still not as widespread as its use in other industries (Lu and Chen, 2022; Zocco et al., 2022). Most of the current operations in recycling plants are performed

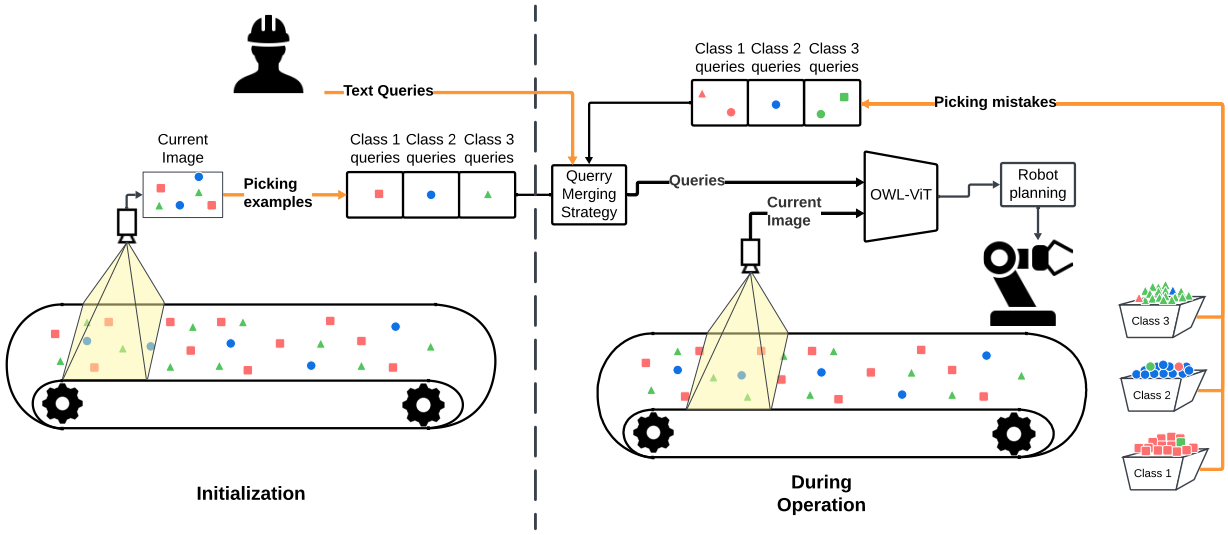


Figure 1: The proposed sorting framework using the active involvement of an operator for initializing the few-shot detector and then continually improving the performance by introducing the sorting mistakes.

by manual labor (van Duijn et al., 2022; European Environment Agency, n.d.), nonetheless, the shortage of the human labor force in this industry is concerning (McGrath, 2021; Resource In Focus, 2020). In addition, the throughput of the current man-powered plants is not sufficient for the ever-increasing volume of waste being produced. Due to these difficulties, any level of automation in waste management plants can amplify outputs and contribute to visions of a circular economy.

The policies of the circular economy dictate different treatments for waste, depending on its type, value, and conditions. Specifically for electronic waste (e-waste), correct assessment is necessary to sort the items so that they get processed differently. Possible treatments can be reusing items entirely or in part, refurbishing, dismantling to recover the materials, and, as a last resort, disposing of the waste. Furthermore, correct separation between waste items based on their content is desirable to increase the purity of material recovery. Hence, one of the most challenging tasks in e-waste management is automated and accurate sorting of the waste.

Many forms of AI models can be deployed in the sorting process. Vision-based models have been the most appealing option (Maier et al., 2024) since they only rely on a

camera sensor to observe and infer about objects, making them the cheapest and most easily available sensors. However, the quality of vision-based sorting is determined by the deep learning vision model behind the scenes that is processing the images. Deep *Object Detection* models are usually the most common form of vision models for this task (Li et al., 2019; Rastari et al., 2024; Zajec et al., 2024). For every image taken as input, they can locate different items and determine their class. This will be enough for downstream robotic actuators to displace and sort the items physically. Models like Faster R-CNN (Ren et al., 2016), SSD (Liu et al., 2015), and YOLO (Redmon et al., 2015), given adequate training, have proven effective in previous studies and applications. Nevertheless, training deep learning models requires vast amounts of labeled data and expensive computational resources, making the process time-consuming and costly. Moreover, the traditional object detectors are limited to the knowledge they gain from the training data and only during the training process. If the user wants to make any change to their performance, such as introducing novel classes, adding new examples for existing classes, correcting the mistakes of the model, or emphasizing a certain class, they should go back through the data collection and model training again. As a result, relying on these models can be difficult for situations where rapid deployment of custom detection tasks is essential, and the reliance on machine learning expertise makes the current workflows difficult to follow, especially for Small and Medium-Sized Enterprises (SMEs).

In recent years, the advancement of the *Transformer* models has opened up new opportunities in the field of object detection tasks. Notably, models such as *Open World Localization Vision Transformers* (OWL-ViT) (Minderer et al., 2023) have pushed the boundaries by utilizing text and image embeddings, enabling them to identify objects with minimal supervision. The framework that we present, in order to leverage the new possibilities provided by open-world models, is illustrated in Figure 1. In this framework, the deployment of OWL-ViT in a sorting scenario is divided into two phases of initialization and during operation. In the initialization, instead of collecting numerous examples for each class of e-waste, the computer-science-unaware operator initializes the model with only one query for each class. The query can be a text description of each item or an example image of an

item belonging to that class. The model can also be adjusted during the operation, when the mistakes made by the model can be picked and introduced to the model through image queries. Then, the initial and during-operation queries can be merged through methods that we discuss later in this work. The output of merging will influence the performance of the OWL-ViT model such that it constantly adjusts itself to conform to the operator's needs.

The contributions of this work can be summarized as: **I**) To the best of the authors' knowledge, this research is the first to apply the concept of few-shot object detection for vision-based sorting of e-waste. **II**) We extend the OWL-ViT v2 model to perform few-shot detection. To this end, a few techniques for merging queries will be described and compared. **III**) We compare conventional closed-set object detection models with our few-shot model in terms of training time, required data, inference time, accuracy, and continual learning ability, highlighting the few-shot approach's balanced performance and overall lead. **IV**) We provide an open-source implementation of our method in the ROS2 framework for easy implementation of few-shot learning for robotic applications.

The outline of the article is as follows: Section 2 reviews some of the related works. In section 3, we discuss our methodology in more detail. Section 4 presents the results of our approach and draws a comparison with traditional object detection. Finally, section 5 concludes the article and discusses potential future directions for expanding this research.

## 2. Related Work

In this section, in search of the desired adaptability and plasticity for the sorting AI model, we have explored some of the state-of-the-art deep learning practices that are partly or entirely relevant to our research goals.

### *2.1. Active Learning and Human-in-the-Loop Systems*

The main goal of *Active Learning* (AL) is to maximize a model's performance, exploiting a minimal set of annotated samples, which is given prior to training or by online interaction with a human expert (Ren et al., 2021; Cacciarelli and Kulahci, 2024). Different strategies are followed to select samples from the dataset, for human experts to annotate,

including uncertainty-based and diversity-based (Ren et al., 2021). The simplest approach is uncertainty-based, where the model is used to classify all samples, and based on the confidence of outputs, a decision is made about whether each sample should be selected for annotation or not (Scheffer et al., 2001; Settles, 2009). Diversity-based AL aims to pick samples that best represent the dataset's distribution (Sinha et al., 2019; Sener and Savarese, 2017). For instance, in (Sener and Savarese, 2017), a core-set of the dataset is calculated such that training on this core-set is equivalent to training on the entire dataset.

Deploying pretrained models is another time and resource-efficient approach to have active involvement of a human expert in the model training/improvement (Wang et al., 2022a; Smith et al., 2023). In this approach, instead of having the expert work on the dataset and improve the quality of the labeling, they can directly instruct the model and adjust its performance during deployment. This approach, despite previously mentioned AL, is more aligned with the no-training requirements and brings model improvements on the go.

### *2.2. Continual Learning*

Continual learning focuses on developing intelligent systems that can accumulate or update knowledge over time (Hadsell et al., 2020; Parisi et al., 2019). The goal is to preserve the learned knowledge even if the model is trained on a completely new dataset, overcoming the phenomenon of catastrophic forgetting. Continual learning is applicable to scenarios where the AI model faces new tasks, new examples of old tasks, or a new environment(Wang et al., 2024).

Forming replay buffers from old and new tasks, or adding regularization terms to the loss function, where common approaches to continual learning (Wang et al., 2024). Recently, rehearsal-free continual learning aims to eliminate revisiting previous tasks and training data. The idea is that the task to be done by the AI model is described through a prompt, and the model easily adapts to the new task (Wang et al., 2022b; Smith et al., 2023). For instance, (Wang et al., 2022c) dynamically generates prompts for a pre-trained model upon any modifications to the task or label space. To face the challenge of an unknown task

during the test time, (Wang et al., 2022a) matches the upcoming sample embedding with key-value pairs of previously saved task-specific prompts and picks the closest task prompt. Some of the mentioned methods propose an object classification pipeline, built on top of vision transformers. However, newer pre-trained transformer models (Oquab et al., 2023; Minderer et al., 2022) with object detection and segmentation capabilities are introduced that can be adopted for continual learning.

### *2.3. Open World Detection*

After the impressive advances in *Large Language Models* (LLMs) performance and scalability, the field of computer vision also benefited from *Vision Transformer* (ViT) models (Dosovitskiy, 2020; Carion et al., 2020; Liu et al., 2021). Most object detection models, including those based on ViT, are trained on a fixed set of classes, which limits them to recognizing only the objects they have seen during training. Consequently, they do not naturally generalize to unseen classes or support incremental learning from new examples without significant re-training (Zhu and Chen, 2024). Hence, the open-world detection models were introduced that generally store knowledge about a far greater number of classes and can be interacted with using textual or visual prompts (Li et al., 2024; Minderer et al., 2022; Liu et al., 2025; Zou et al., 2024). Initial attempts provided models that could detect/segment objects introduced through a text prompt (Liu et al., 2025). However, in many real-world applications, the objects to be detected are difficult to describe through text, hence, a visual example of the target object can serve similar to a text prompt (Minderer et al., 2022). Furthermore, in (Li et al., 2024), the concept of in-context prompts was introduced, extending the number of visual prompts per class (in-context examples) to more than one.

### *2.4. Adaptable AI in Waste Sorting*

The idea of using flexible object detection models in an industrial setting is not new, and previous attempts have tried to address it. For instance, RILOD (Li et al., 2019) implements a model with continual learning capabilities to edge devices, which are essential for industrial implementation of object detection. In (Trajkova et al., 2021; Rožanec et al., 2022), active

| Reference | Task | Model | Active | Continual | No Pre-training | Application | Prompt Type | Merging Prompts |
|---|---|---|---|---|---|---|---|---|
| (Rožanec et al., 2022) | Classification | ResNet-18 | ✓ | ✗ | ✗ | Quality Control | ✗ | ✗ |
| (Rastari et al., 2024) | Object Detection | YOLO v8 | ✗ | ✗ | ✗ | Waste | ✗ | ✗ |
| (Zajec et al., 2024) | Classification | ViT+ProtoNet | ✓ | ✗ | ✓ | Quality Control | Image | ✗ |
| (Sirimewan et al., 2024) | Segmentation | DouSeg++ | ✗ | ✗ | ✗ | Construction Waste | ✗ | ✗ |
| (Tur et al., 2024) | Classification | CLIP, DINOv2 | ✓ | ✓ | ✓ | Retail | Text, Image | Averaging |
| (Babé et al., 2025) | Classification | CLIP, DINOv2 | ✗ | ✗ | ✗ | Waste | ✗ | ✗ |
| (Ranjbar et al., 2025) | Classification | EVA-CLIP | ✓ | ✓ | ✓ | Plastic Waste | Text, Image | Averaging |
| **Our work** | **Object Detection** | **OWL-ViT v2** | ✓ | ✓ | ✓ | **E-Waste** | **Text, Image** | Fine-Grained |

Table 1: A summary of the most related research to our work.

learning techniques were deployed in industrial inspection to minimize data collection and annotation without performance compromise. Moreover, (Zajec et al., 2024) used few-shot learning for defect detection in manufacturing.

In the specific case of sorting, especially in recycling industries, most of the existing technologies take advantage of closed-set object detection such as YOLO models (Rastari et al., 2024). A semi-supervised model was proposed by (Sirimewan et al., 2024) for the segmentation of construction waste images, trying to minimize reliance on a large annotated dataset for training. The work of (Babé et al., 2025) investigated the generalization power of foundation models as backbones for vision-based waste sorting. In the works most similar to our research, (Ranjbar et al., 2025; Tur et al., 2024) deploy zero/few-shot vision-language models respectively for tasks of waste and retail sorting. However, they only considered query averaging for merging multiple prompts per class, and their model is not designed to help the actuation of waste items by localizing them in the image. Table 1 summarizes the new aspect of our work with respect to the literature.

Our research is built on the junction of continual learning and active learning. The similarity of the present work to the field of continual learning lies in the fact that we are dealing with a stream of examples being added to our dataset, some of which fall outside the predefined set of classes. As a result, the task defined for the model is continuously evolving. The fact that our model is trained by fetching a minimal subset of data from the real-time stream of images and using a human expert's judgment for labeling this subset defines the active learning aspect of our research.

## 3. Methodology

The proposed methodology addresses the active-continual-learning problem of sorting by utilizing a pre-trained vision-language transformer named OWL-ViT v2. This model performs object detection based on the input image or text embeddings that it receives. This feature makes it possible to manipulate and adjust input embeddings to get the desired behavior from the model. In this section, the baseline model is reviewed, our embedding merging strategies are discussed, and alternative transfer learning approaches are introduced.

### *3.1. Overview of OWL-ViT-v2 Model*

Since the introduction of the *Contrastive Language-Image Pretraining* (CLIP) (Radford et al., 2021), the performance of vision-language models has improved significantly, and it has been possible to pre-train relatively large ViT models on vast datasets. The OWL-ViT exhibits the same pre-training scheme as CLIP. At the bottom of Figure 2, an overall view of the modules inside OWL-ViT v2 is presented. The backbone of this network is a vision transformer that is trained contrastively, similar to CLIP. Despite the traditional classification training that only presents correspondences between images and their respective text during training, contrastive learning presents every image with one true text correspondence and several false correspondences, helping the model to establish scalable knowledge about tasks.

From all the modules inside OWL-ViT v2, only *Backbone*, *Text Embedder*, and *Image Embedder* were trained contrastively on large datasets in a loosely supervised manner. The

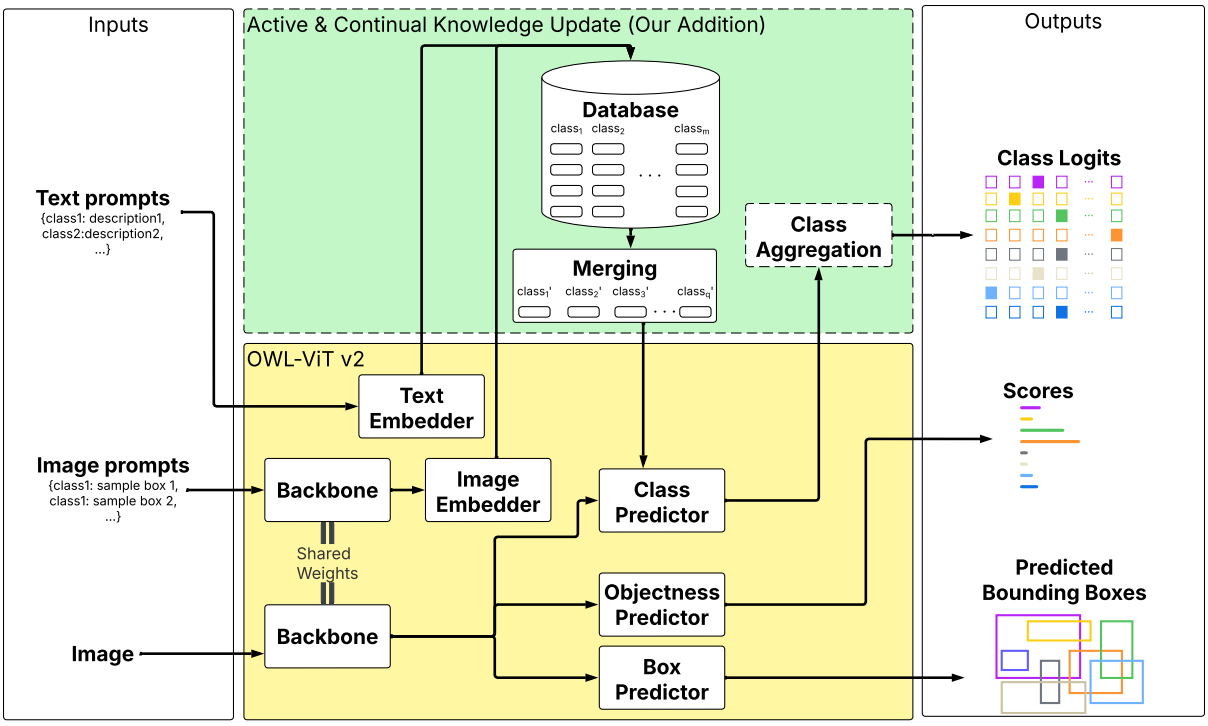


Figure 2: Overview of the proposed architecture, illustrating how the expert supplies text and image prompts to a continually updated knowledge database, which is then merged with the OWL-ViT v2 pipeline. The pipeline comprises a backbone, text and image embedders, as well as objectness, box, and class predictors, enabling flexible and continual visual detection.

*Objectness Predictor* and *Box Predictors* were separately trained on closed-set datasets dedicated to object detection, while keeping the backbone frozen. The *Class Predictor* in the vanilla OWL-ViT accepts a prompt (either the output of the text embedder or the image embedder) and the feature map calculated by the backbone, while in our version portrayed in Figure 2, the prompt is manipulated for active continual learning capabilities. The prompt consists of any number of arbitrary class names and a single corresponding embedding vector for each class.

If each embedding vector outputted by image embedder or text embedder is denoted by $\mathbf{v} \in \mathbb{R}^{512}$, and if we intend to detect $m$ number of classes, one prompt would be a two dimensional matrix $P \in \mathbb{R}^{m \times 512}$. Moreover, the class predictor serves as a function $\mathscr{F}(P, M) \rightarrow L$ where the $M$ is the feature map coming from the backbone and $L \in \mathbb{R}^{3600 \times m}$

is the logits matrix. The class predictor $\mathscr{F}$ of the OWL-ViT model is flexible about the number of input classes $m$ such that the model can classify over any arbitrary number of classes introduced through the $P$.

In order to achieve zero-shot/text-conditioned detection, the user must provide one text description of each of the desired classes. Moreover, to achieve one-shot/image-conditioned detection, one image example of each of the target classes must be provided. Image-conditioned is especially useful when describing the target objects is difficult or not representative of their appearance.

### *3.2. Active Continual Learning by Multiple Image Embedding per Class*

Despite facilitating active learning by minimizing the required annotation per class to only one, OWL-ViT struggles in continual learning, where further samples of individual classes are necessary to cover different aspects of one class. As illustrated in Figure 2, all the text or image queries of the human expert are maintained in a database. In a continual learning problem, the expert is able to identify if a sample is wrongly classified, then he adds this wrongly classified sample to the list of its true class samples, hence the number of samples per class might be more than one. To interface these embeddings with the $\mathscr{F}$, a mapping $\mathscr{M}$ from each class's list of embeddings $E$ to the final embedding $\mathbf{v}$ of that class is required, which is given by

$$\mathbf{v}_i = \mathscr{M}(E_i), \quad E_i = \{\mathbf{v}_{i_j}\}_{j=1}^{k_i}, \tag{1}$$

where $E_i, i = 1, \ldots, m$ is the list of embedding saved for each class, $k_i$ is the number of samples we have for that class, and $\mathbf{v}_i$ is the final embedding of each class that will constitute $P = [\mathbf{v}_1^\top, ..., \mathbf{v}_m^\top]^\top$, which is the input to the class predictor.

#### *3.2.1. Averaging*

Although the OWL-ViT has mainly been designed for one-shot object detection, the authors have proposed averaging further examples of each individual class to achieve few-shot capability (Minderer et al., 2022). Hence, the mapping would be

$$\mathscr{M}(E_i) = \frac{1}{k_i} \sum_{j=1}^{k_i} \mathbf{v}_{i_j}. \tag{2}$$

An alternative to averaging could be geometric median, which is less sensitive to noise and outliers. The mapping $\mathscr{M}$ will be

$$\mathscr{M}(E_i) = \arg\min_{\mathbf{z}} \sum_{j=1}^{k_i} \|\mathbf{v}_{i_j} - \mathbf{z}\|. \tag{3}$$

*3.2.2. Fine-Graining*

To find the average or median of all embedding samples for every class is a relatively naive approach to the problem of enabling few-shot learning for OWL-ViT, since it can lead to a loss of information and variability in class samples. As an alternative, each sample in a class can be treated as a separate subclass, and the prompt matrix will have an extended shape of $P \in \mathbb{R}^{q \times 512}$ where $q = \sum_{i=1}^{m} k_i$. This merging strategy is also clear in Figure 2, where the original classes ($\text{class}_1$, $\text{class}_1$, ..., $\text{class}_\text{m}$) are transformed into fine-grained intermediate classes ($\text{class'}_1$, $\text{class'}_1$, ..., $\text{class'}_\text{q}$). Note that if the fine-graining is used, a *Class Aggregation* step after prediction is needed to invert from fine-grained classes back to the original class space.

*3.2.3. Clustering*

Fine-graining can potentially increase the size of the prompt matrix to an unbounded value, depending on the variability in each class, which can affect the computations in the class predictor. An alternative could be to cluster the embeddings of each class into sub-categories and find the average or median in each of the categories. In this case, the prompt matrix will be $P \in \mathbb{R}^{r \times 512}$ where $m < r < q$. We have used K-Means for this purpose to create smaller yet representative divisions inside each class's embeddings.

*3.2.4. Transfer Learning*

The previous query manipulations were mainly focused on creating a prompt matrix with dimensions of size $m \times 512$, so that the workflow of the OWL-ViT remains intact. A

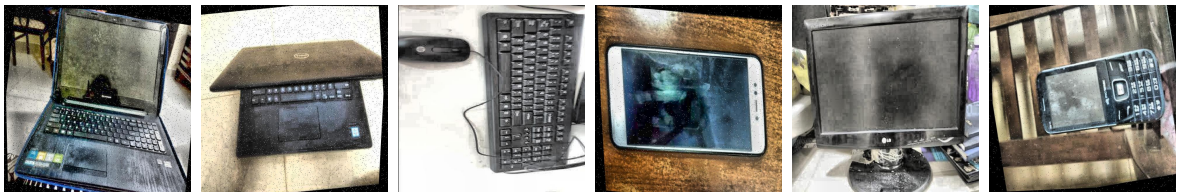

Figure 3: Sample images of the e-waste dataset used for this study.

major bottleneck, preventing few-shot capability in OWL-ViT, is the class predictor, which currently only accepts one embedding vector per class. Another approach to mitigate this issue is to replace the class predictor with a closed-set classifier that will require retraining upon any changes in the embedding database.

Although this is a setback from the original training-free adaptability of the solution, one of the important requirements of this research, in case the classifier is lightweight and generalizable enough to be trained in a very short time, can still be used in the context of active continual learning. Due to the fact that the embedding vectors outputted by the image embedder are equivalent to the embeddings from text embedder, it can be intuitively claimed that these embeddings serve as a textual description of any object, thus they are very much generalizable and descriptive, and performing classification based on them should also be easily generalizable. To this end, we have experimented replacing the class predictor with a *Multi-Layer Perceptron* (MLP) and *K-Nearest Neighbors* (KNN) classifiers. The MLP is constructed from two hidden fully connected layers of shapes 128 and 64. Both layers use the ReLU activation function. The final output layer maps the hidden representation to the fixed number of classes. The cross-entropy loss function is applied during training, and the model is trained using the Adam optimizer with a learning rate of 0.001.

## 4. Result and Discussion

As a case study, a public annotated dataset(Dataset, 2024) consisting of electronic waste items was used for evaluation. More than 4060 images of adapters, chargers, keyboards, laptops, memory, mobile phones, monitors, and mouses are annotated, and Figure 3 shows a few sample images from this data.

| Model | Task | Merging | Cls. Accuracy (%) | mAP50 | Training time (s) | Inference time (s) | $k$ |
|---|---|---|---|---|---|---|---|
| YOLOv11-$l$ | - | - | 85.78 | 0.919 | 7120 | 0.094 | 3564 |
| RT-DETR-$l$ | - | - | 87.34 | 0.956 | 9710 | 0.091 | 3564 |
| Faster-RCNN-*ResNet50* | - | - | 80.28 | 0.887 | 9024 | 0.027 | 3564 |
| ViT-B16 + MLP | - | - | 96.12 | - | 1094 | 0.258 | 3564 |
| ViT-B16 + KNN | - | - | 95.26 | - | 1089 | 0.287 | 3564 |
| OWL-ViT v2 | zero-shot | - | 63.47 | 0.418 | 0.0 | 0.257 | 0 |
| | few-shot | average | 60.89 | 0.399 | 0.0 | 0.252 | 10 |
| | | median | 61.83 | 0.413 | 0.0 | 0.255 | 10 |
| | | fine-grained | 58.31 | 0.405 | 0.0 | 0.260 | 10 |
| | | fine-grained clustered | 61.12 | 0.427 | 0.0 | 0.279 | 10 |

Table 2: Performance comparison between proposed active continual learning model for e-waste sorting and state-of-the-art conventional closed-set learning methods. The presented metrics are classification accuracy, mean average precision, training time, and run time for each image.

For comparison, 3 popular closed-set object detection methods, including state-of-the-art YOLOv11-$l$ (Jocher and Qiu, 2024) (here $l$ stands for large configuration of the network), RT-DETR-$l$ (Lv et al., 2023), and Faster-RCNN (Ren et al., 2016) with ResNet50 backbone were trained. All proposed configurations of OWL-ViT v2 for few-shot learning were also benchmarked on the same dataset, and the comparative results are gathered in Table 2. Note that the methods named as ViT-B16+*classifier* are referring to using the backbone and image embedder of OWL-ViT and replacing the class predictor with a custom classifier. As discussed in section 3.2, the new classifiers will not have any zero/one/few-shot capabilities, but they will operate on highly generalizable embeddings from OWL-ViT. Classification accuracy is presented in Table 2 to solely show the classification power of methods without taking the bounding-box prediction aspect of models. To measure the classification accuracy, the number of predictions with an IoU of more than 0.5 with the ground-truth box and confidence above 0.4 was considered as correct classifications. Other metrics considered in Table 2 are the number of training samples it took ($k$) and the training time spent to get the models to the reported accuracy, as well as the time it takes to perform inference on a single image. The model training reported in this paper was conducted on a system configured

with 32GB of RAM, an Intel Core i9-13900H CPU, and an NVIDIA GeForce RTX 4090 Laptop GPU running CUDA 12.5.

Among the investigated methods, ViT-B16 (Radford et al., 2021)+MLP and RT-DETR-*l* respectively yielded the highest classification accuracy and highest average precision (mAP50). Generally, closed-set methods (YOLOv11-*l*, RT-DETR-*l*, Faster-RCNN-*ResNet50*, ViT-B16+MLP, and ViT-B16+KNN) show a superior accuracy and mAP in our benchmark, while they consume a high number of training samples (parameter $k$ in Table 2) to achieve this state. However, the transfer learning approaches built on top of OWL-ViT's image embedder (ViT-B16+MLP, and ViT-B16+KNN) take far less training time, since they only train a small classifier added to the OWL-ViT and the majority of the OWL-ViT model remains intact during the training. The mAP50 metric was not reported for these two classification heads since they do not have any bounding box prediction capability and are merely classifiers that rely on OWL-ViT's objectness predictor to locate the object.

Unlike the closed-set methods, OWL-ViT-derived approaches have zero domain-specific training and take far fewer samples to learn the task, hence, they can be actively trained on the implementation site. Despite low training time and low sample requirements, these methods suffer from higher runtime compared to traditional object detectors.

The vanilla OWL-ViT v2 with text prompts (zero-shot detection) showed the highest accuracy on all objects for $k = 10$. However, by looking at recall per each class, presented in Table 3, the zero-shot method had good performance for more familiar class descriptions, such as keyboard, monitor, and mouse, while having a bad performance on vague class descriptions such as adapter and memory. This indicates that the familiar classes may had been better represented in the extensive pre-training of OWL-ViT. To address failures in adapter and memory detection, a better textual description of these classes can be helpful, but determining a good text description of a class is often not straightforward. Here, image-conditioned detection came into play, such that instead of using language to describe the target class, one or more image examples were given. From different investigated strategies for merging the image prompts, finding the geometric median had the most promising accuracy of 61.83%. The given image samples increase the accuracy in the adapter and memory

| Model | Task | Merging | adapter (%) | keyboard (%) | laptop (%) | memory (%) | mobile (%) | monitor (%) | mouse (%) |
|---|---|---|---|---|---|---|---|---|---|
| YOLOv11-*l* | - | - | 100.0 | 88.8 | 93.7 | 90.9 | 93.9 | 93.4 | 86.0 |
| RT-DETR-*l* | - | - | 100.0 | 92.4 | 94.1 | 100.0 | 95.3 | 88.0 | 75.4 |
| Faster-RCNN-*ResNet50* | - | - | 80.0 | 88.4 | 96.6 | 81.8 | 96.5 | 95.6 | 90.0 |
| ViT-B16 + MLP | - | - | 100.0 | 93.5 | 95.8 | 100.0 | 96.2 | 100.0 | 94.7 |
| ViT-B16 + KNN | - | - | 75.0 | 95.7 | 90.5 | 100.0 | 98.5 | 96.9 | 94.7 |
| OWL-ViT v2 | zero-shot | - | 0.0 | 78.8 | 51.69 | 0.0 | 66.4 | 76.0 | 74.0 |
| | few-shot | average | 25.0 | 78.8 | 39.8 | 9.0 | 77.4 | 43.4 | 74.0 |
| | | median | 25.0 | 78.8 | 37.2 | 9.0 | 77.4 | 58.7 | 74.0 |
| | | fine-grained | 25.0 | 76.9 | 36.4 | 9.0 | 63.7 | 76.0 | 72.0 |
| | | fine-grained clustered | 25.0 | 76.9 | 42.3 | 9.0 | 66.4 | 76.0 | 74.0 |

Table 3: Comparison of recall per class for all investigated methods. Similar to Table 2, the $k$ is equal to 10 for all few-shot approaches.

classes, where the zero-shot method completely failed.

In tables 2 and 3, the metrics of the few-shot method are presented under the condition $k = 10$, indicating that 10 samples were provided to the model as prompts. However, it is important to emphasize that the total number of samples per class exerts a significant influence on the resulting accuracy.

Figure 4 compares different merging strategies for few-shot image-conditioned learning. As is evident in the figure, when the number of samples grows for each class, the classification accuracy grows as well for all merging strategies. This positive correlation indicates that the few-shot learning can learn from active feedback of the operator during deployment. Fine-grained merging shows the highest increase among methods, starting with 41.22% accuracy for $k = 1$ and achieving 66.51% after the addition of $k = 80$ samples for each class. Observing the results from Table 2 and Figure 4, we selected the fine-grained merging strategy as our main method, since it has close performance metrics to the other query merging approaches, but yields better learning from new samples among the batch of investigated methods.

Figure 5, summarizes five key criteria: training time, required sample size, accuracy, inference speed, and adaptive accuracy improvement during deployment with expert corrections. Each metric is normalized (denoted by an overbar) in this web chart. Our few-shot

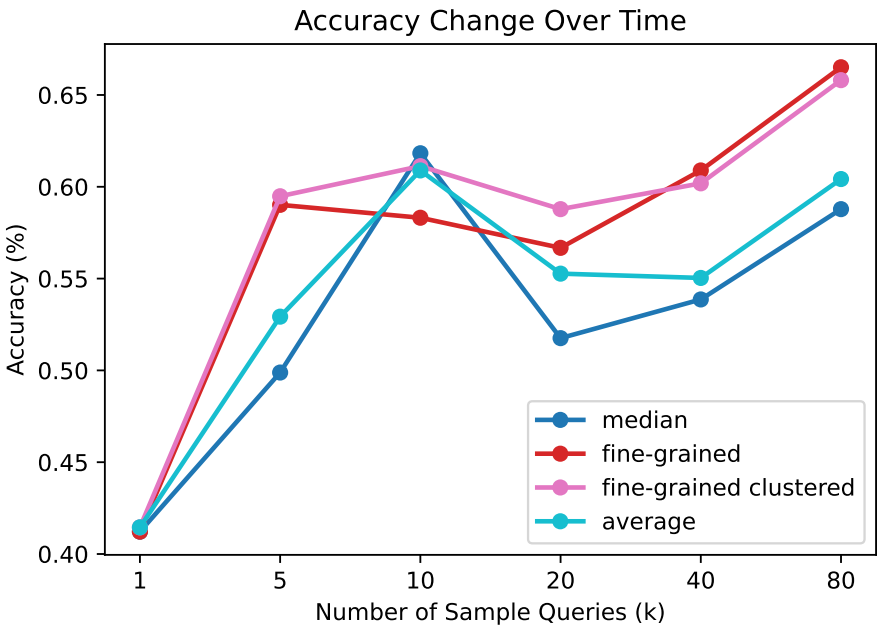


Figure 4: This figure demonstrates a positive correlation between model accuracy and the number of samples given for each class.

approach, based on OWL-ViT v2 with fine-grained query merging, requires the least training time and data, and is able to improve its accuracy during deployment, making it the most adaptable and interactive.

## 5. Conclusion

This study introduces an active-continual-learning approach for e-waste sorting, evaluated against several state-of-the-art closed-set learning methods. The proposed solution is not limited to e-waste but can also be applied to various sorting applications. As shown in Tables I and II, our framework achieves competitive classification accuracy while requiring minimal training time and a smaller dataset. Additionally, the few-shot learning paradigm enables effective adaptation to newly introduced samples, resulting in flexible performance during deployment and interactive operator-driven improvements. These findings underscore the method's ability to operate in dynamic, real-world settings where categories may be incrementally added over time.

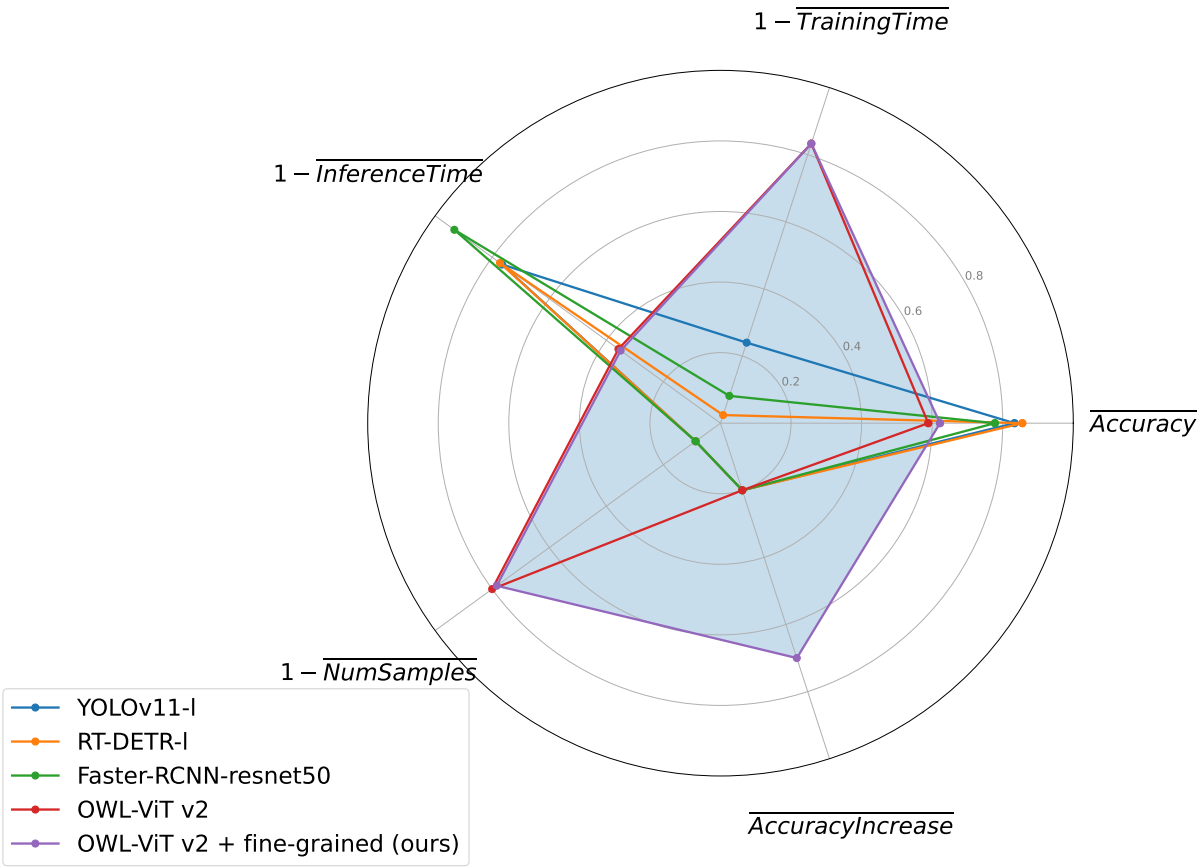


Figure 5: Comparison of the proposed few-shot learning approach with traditional deep learning methods commonly used in sorting.

Looking ahead, in our future works, we hope to address some of the remaining challenges in deploying adaptable vision-language models in industrial settings. Although our training-free model based on OWL-ViT v2 offered several benefits, there remains room for accuracy improvements to surpass the level of specialized models. Moreover, as the field of vision-language models is constantly evolving, new architectures should be explored that support multi-sample per class without requiring query manipulation. Real-time deployment of the proposed approach on edge devices will also be challenging due to a relatively low update rate; therefore, we plan to investigate the integration of efficient single-object tracking to achieve real-time processing. By building on the insights gained in this study, we aim to advance sustainable e-waste management through more accurate, efficient, and flexible

vision-based solutions.

## Statements and Declarations

### *5.1. Declaration of competing interest*

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### *5.2. Code availability*

The developed source code used to generate all results in this paper will be made publicly available on GitHub upon acceptance of the manuscript.

### *5.3. Declaration of Generative AI and AI-assisted technologies in the writing process*

During the preparation of this work, the authors used ChatGPT o3-mini in order to improve readability and language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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